Literature Review: Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video

The transition of urban traffic management from static, infrastructure-heavy sensor arrays to dynamic, visual intelligence systems represents a critical evolution in intelligent transportation systems (ITS). The deployment of roadside closed-circuit television (CCTV) cameras offers an extraordinarily rich data modality. However, translating raw pixel data into reliable vehicle counts and subsequent short-term congestion forecasts involves navigating a gauntlet of computational and environmental challenges. This literature review synthesizes the state-of-the-art research required to fulfill the engineering specifications outlined in the DSCI-41 Capstone Project: Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video[1].

The DSCI-41 specification mandates a rigorous, multi-stage engineering pipeline. The baseline requirement, defined as the O2 reference, necessitates handling mixed vehicle types and oblique perspectives. The primary technical contribution, defined as the O3 objective, requires sustaining counting accuracy under severe visual degradation, including occlusion, rain, and low light. Finally, the O4 objective mandates the integration of these volume estimates into a complete operable system capable of forecasting near-term congestion, culminating in an independent acceptance phase (O5) governed by strict Key Performance Indicators (KPIs) and negative testing campaigns[1]. By critically analyzing recent advancements in deep learning-based object detection, multi-object tracking, geometric camera calibration, and spatio-temporal graph forecasting models, this report delineates the theoretical foundations and practical deployments necessary to construct and validate this visual intelligence system.

1. Establishing the Baseline: Mixed Vehicle Types and Oblique Perspectives

The foundational step in estimating traffic volume is the accurate, real-time detection of vehicles within a video frame. To establish a reproducible baseline capable of meeting the DSCI-41 O2 reference, the system must robustly identify a heterogeneous mix of vehicle classes under the geometric distortions inherent to roadside camera placements.

1.1 Culturally Adaptive Detection in Heterogeneous Traffic

Traffic volume estimation models trained on structured Western datasets frequently suffer catastrophic accuracy degradation when deployed in developing regions or unstructured environments. The literature emphasizes the necessity of "cultural adaptation" in visual models to handle mixed vehicle types, which is a core requirement of the O2 baseline[1]. Datasets such as the Indian Driving Dataset (IDD) and DriveIndia reveal that unstructured traffic environments contain a diverse mix of vehicles, including auto-rickshaws, animal-drawn carts, and densely

packed two-wheelers moving without strict lane discipline[3].
The IDD captures this complexity across 34 distinct classes, highlighting a severe "long-tail" distribution where dominant classes like cars and motorcycles vastly outnumber rare classes such as animals or specialized carts[2]. Similarly, the DriveIndia dataset provides nearly 67,000 high-resolution images across 24 traffic-relevant categories, utilizing a strict bounding box policy where occluded objects are annotated only for their visible portions[4]. Training models solely on these localized datasets, however, can lead to environmental overfitting. A highly effective strategy identified in recent research is data fusion. By combining structurally complex datasets like IDD with environmentally diverse datasets like BDD100K (Berkeley DeepDrive)—which contains extreme weather variations—researchers have trained hybrid models that are simultaneously culturally adaptive to unstructured vehicle mixes and environmentally robust to adverse weather conditions[2]. For example, customized semantic segmentation architectures like IndiVNet have achieved a Mean Intersection over Union (mIoU) of 69.98% on the IDD by incorporating extensive dilution techniques that broaden the field of view to capture rich contextual information[5].

1.2 Geometric Calibration and Inverse Perspective Mapping

A critical vulnerability in raw roadside video analysis is the oblique perspective. Cameras mounted on roadside poles look down at traffic at an angle, causing perspective distortion where a vehicle located near the camera occupies far more pixels than a physically identical vehicle further away. Without geometric correction, spatial relationships, distances, and speeds extracted from the two-dimensional image plane are inherently distorted, compromising the accuracy of the O2 reference system[6].
To translate two-dimensional pixel coordinates into three-dimensional real-world coordinates, the system must undergo camera calibration to extract intrinsic and extrinsic parameters. In traffic surveillance, manual calibration is rarely feasible. Instead, researchers rely on auto-calibration utilizing the geometric properties of the scene. By identifying parallel lane lines and orthogonal structures, the system can calculate vanishing points that define the ground plane relative to the camera[8]. Once calibrated, a homography matrix is computed. Inverse Perspective Mapping (IPM) uses this matrix to project the oblique image onto a top-down, Bird's-Eye View (BEV). In this transformed space, perspective distortion is theoretically eliminated, and the Euclidean distance between any two transformed points accurately reflects their true physical separation on the road surface[6].

3.3 The Planar Assumption and 3D Volumetric Alternatives

A profound second-order insight derived from the literature is the structural limitation of traditional homography. The mathematical formulation of homography strictly assumes that all points lie on a perfectly flat ground plane. In reality, urban roads possess camber for drainage, gradients, and speed bumps. Furthermore, two-dimensional bounding boxes enclose the entire vehicle, meaning the bottom-center of the box may not accurately represent where the vehicle's tires contact the pavement, especially for vehicles viewed from a sharp oblique angle. When a point off the true ground plane is warped via IPM, it suffers from severe projection

stretching, leading to significant spatial positioning errors[12].
To circumvent this limitation, advanced systems are pivoting toward three-dimensional perception without explicit real-time algorithmic calibration. One approach involves the use of three-dimensional Look-Up Tables (LUTs). By utilizing pre-existing point clouds generated by Mobile Mapping Systems (MMS) or LiDAR, researchers can project the point cloud onto the camera's image plane. This creates a static LUT where every pixel is permanently mapped to an exact spatial coordinate. When a vehicle's ground contact point is detected, the system queries the LUT, bypassing the planar homography assumption entirely and reducing positioning errors by an average of 61.4%[12]. Alternatively, advanced monocular networks are trained to regress complete three-dimensional cuboids, providing volumetric data that allows the system to track the exact centroid of the vehicle in space. This heavily improves speed estimation, accurate vehicle counts, and Time-to-Collision calculations essential for Near-Miss criteria evaluations[6].

Geometric Transformation Method	Foundational Mechanism	Primary Strengths	Limitations and Vulnerabilities
Vanishing Point Calibration	Extracts scene line features to infer perspective parameters automatically.	Requires no manual field measurements; adaptable to existing standard CCTV networks.	Fails in featureless environments; highly sensitive to camera shake or occlusion of lane markings.
Inverse Perspective Mapping (IPM)	Warps images via a homography matrix to create a Bird's-Eye View.	Eliminates perspective distortion for spatial proximity calculations.	Strictly assumes a perfectly flat ground plane; introduces projection stretching for tall vehicles.
3D Look-Up Tables (LUTs)	Maps 2D pixels directly to 3D coordinates using pre-existing Mobile Mapping System point clouds.	Bypasses planar assumptions entirely; reduces positioning errors by over 60%.	Requires high-quality, pre-existing LiDAR or MMS data for the specific surveillance intersection.

2. Engineering the Contribution: Accuracy Under

Occlusion, Rain, and Low Light

The DSCI-41 specification's O3 technical contribution mandates that the engineered system sustain counting accuracy under severe visual degradation, specifically occlusion, rain, and low light[1]. This requires moving beyond baseline object detection to deploy architectures specifically optimized for degraded visual environments.

2.1 Optimization of Single-Stage Detectors

The literature heavily favors deep Convolutional Neural Networks (CNNs) for vehicle detection. Two-stage detectors, such as Faster R-CNN and Mask R-CNN, generate region proposals before performing classification and bounding box regression[13]. While these models historically achieved superior localization accuracy, their high computational latency renders them impractical for real-time edge deployment on roadside cameras where the DSCI-41 KPI-6 demands a strict inference latency threshold (p95 latency less than or equal to 80% of the baseline)[1]. Consequently, single-stage detectors—most notably the YOLO family—have become the prevailing standard for traffic surveillance.

Recent iterations of YOLO have introduced profound architectural innovations specifically beneficial to traffic monitoring under adverse conditions. YOLOv7x, for instance, has demonstrated robust baseline performance with a Mean Average Precision (mAP@0.5) of 0.708 at nearly 88 frames per second, proving its viability for real-time applications[16]. YOLOv8 and YOLOv9 introduced anchor-free detection and programmable gradient information, which significantly improve the network's ability to detect small objects by preserving spatial features deeper into the network[13]. To directly address occlusion, researchers have developed customized variants such as MSN-YOLOv8n. This architecture integrates a C2f_MLCA module and a Detect_SEAM module alongside a Normalized Wasserstein Distance (NWD) loss function. These enhancements improve cross-scale information processing, allowing the network to better detect small or heavily occluded vehicles that traditional IoU-based loss functions would miss[13].

The latest iteration, YOLOv11, heavily optimizes the traditional network architecture for complex traffic scenes. It replaces older modules with C3k2 blocks for lightweight multi-scale feature extraction and introduces the C2PSA spatial attention module. By utilizing spatial pooling combined with dual-branch attention, the model forces the network to focus on salient regions of vehicles, such as windshields or headlights, rather than background distractors[15]. This is critical in low-light environments. Furthermore, YOLOv10 and YOLOv11 emphasize end-to-end Non-Maximum Suppression (NMS)-free training. In dense traffic where vehicles are tightly packed, traditional NMS often erroneously suppresses valid bounding boxes. NMS-free architectures mitigate this algorithmic occlusion, ensuring accurate counts in bumper-to-bumper traffic scenarios[18].

2.2 Transformer-Based Architectures and Motion Enhancement

While CNNs are adept at extracting local features, their restricted receptive fields can struggle with global context. Vision Transformers (ViTs) and models like the Real-Time Detection

Transformer (RT-DETR) bypass this limitation by relying on self-attention mechanisms[14]. RT-DETR processes images via an efficient hybrid encoder and utilizes IoU-aware query selection. The literature indicates that RT-DETR handles scale variations and partial occlusions significantly better than traditional convolutional architectures. In scenarios where a large vehicle obscures smaller vehicles, the global receptive field of the transformer allows the network to infer the presence of the occluded vehicle from disjointed visible parts[19].

To further isolate dynamic traffic from static background objects or parked vehicles, researchers have integrated Transformer-Enhanced Optical Flow (TE-OF) mechanisms. Optical flow vectors compute the displacement of pixels between consecutive frames. By establishing an adaptive threshold on flow magnitude, stationary vehicles can be algorithmically subtracted from the total count. When these optical flow motion masks are cross-validated with network-generated bounding boxes, the system ensures that only dynamically moving objects are registered, yielding a highly robust estimation of active traffic volume[14].

2.3 Environmental Degradation and Low-Light Adaptation

Adverse weather (rain, fog) and low-light conditions fundamentally degrade pixel-level fidelity. Rain introduces dynamic noise and motion blur, while low light strips the image of texture and color, leaving only structural outlines or headlight glare[18]. Traditional background subtraction or rudimentary edge detection fails completely under these conditions.

The literature identifies two distinct pathways to maintain detection accuracy. The first involves image enhancement pre-processing, utilizing techniques like gamma correction, bilateral filtering, or Generative Adversarial Networks (GANs) to map low-light or rainy images to a clear-day domain prior to detection[14]. However, this introduces substantial computational latency, conflicting with the real-time constraints of the DSCI-41 KPI metrics. The second, more efficient pathway involves illumination-aware architectures. Variants such as LLD-YOLO and DRF-YOLO directly integrate illumination-aware modules and attention-enhanced feature extraction within the network's structure. By learning to extract robust structural features independent of contrast—such as automated headlight pairing utilizing Gaussian mixture models—these networks sustain accuracy without the latency penalty of explicit image reconstruction[18].

2.4 Resilient Multi-Object Tracking Algorithms

Detecting a vehicle in a single frame is insufficient for volume estimation; the system must associate that detection across consecutive frames to form a continuous trajectory. This is vital to prevent duplicate counting and to enable speed estimation. The tracking-by-detection paradigm links frame-by-frame bounding boxes using spatial and visual cues[18].

The Simple Online and Realtime Tracker (SORT) relies purely on the Kalman filter for motion prediction and the Hungarian algorithm for data association based on bounding box overlap. While highly computationally efficient, SORT fails catastrophically when vehicles are temporarily occluded, as the purely kinematic model cannot recover lost identities[18]. DeepSORT addresses this by introducing a deep appearance metric, allowing it to re-associate a vehicle using visual texture features after a prolonged occlusion[14].

However, ByteTrack represents a paradigm shift for adverse weather and low-light tracking. Traditional trackers universally discard low-confidence bounding boxes, assuming them to be background noise. In heavy rain or severe occlusion, the detector's confidence for a valid vehicle naturally drops below standard thresholds. ByteTrack addresses this by categorizing detections into high and low confidence. High-confidence boxes are matched first to existing tracks; subsequently, the remaining unmatched tracklets are matched against the low-confidence boxes[16]. This allows the system to maintain trajectories even when vehicles are heavily obscured, resulting in significantly higher Multiple Object Tracking Accuracy (MOTA) and IDF1 scores. When combined with optimized YOLO detectors, ByteTrack has achieved robust tracking metrics (e.g., MOTA 0.289, IDF1 0.725) in highly complex environments, fulfilling the DSCI-41 negative testing requirements for occlusion and weather degradation (NT-1 and NT-2)[1]. To prevent ID carryover during tracking, systems often employ pre-tracking suppression to filter near-duplicate boxes and same-frame duplicate filters to eliminate overlapping track IDs[19].

Tracking Algorithm	Core Association Mechanism	Performance Under Adverse Conditions	Resource Overhead
SORT	Kalman Filter + Bounding Box IoU	Poor; loses track identities instantly upon visual occlusion or dynamic weather interference.	Minimal; excellent for highly constrained edge devices.
DeepSORT	Kinematics + Deep Visual Appearance (Re-ID)	Strong; recovers identities after temporary occlusion by matching visual features.	High; the Re-ID network requires significant GPU compute resources.
ByteTrack	Two-Stage Association utilizing Low-Confidence Detections	Exceptional; sustains trajectories through heavy rain and severe occlusion by preserving low-confidence bounding boxes.	Moderate; relies entirely on the primary detector without needing a separate Re-ID network.

3. System Integration: Short-Term Congestion Forecasting

The O4 objective of the DSCI-41 specification mandates the integration of visual volume estimates into a complete operable system capable of forecasting near-term congestion[1]. Congestion forecasting aims to predict traffic states into the future, which is a highly non-linear problem driven by both local vehicle interactions and global network topologies.

3.1 Spatio-Temporal Graph Convolutional Networks (STGCN)

Traditional forecasting models, such as Autoregressive Integrated Moving Average (ARIMA) or isolated Long Short-Term Memory (LSTM) networks, treat traffic as a one-dimensional time series. The fatal flaw of these models is their inability to process spatial dependencies[24]. Traffic is inherently a network phenomenon; a bottleneck at one intersection will systematically propagate upstream. To capture this, researchers model the road network as a graph, where nodes represent camera locations or intersections, edges represent connecting roads, and an adjacency matrix defines spatial connectivity[24].

Spatio-Temporal Graph Convolutional Networks (STGCN) have emerged as the dominant architecture for mid-to-long term traffic prediction. STGCN abandons recurrent units entirely in favor of fully convolutional structures. This significantly reduces training time and prevents the vanishing gradient problem inherent in long sequence modeling[25]. The STGCN architecture operates on a block structure containing spatial graph convolutions and temporal gated convolutions. To avoid the massive computational cost of spectral eigen-decomposition on large graphs, STGCN utilizes Chebyshev polynomials to approximate the graph Laplacian. This effectively localizes the convolution filter to a specific radius of hops from a central node, capturing immediate upstream and downstream spatial dependencies[24]. Flanking the spatial layer are one-dimensional causal convolutions combined with Gated Linear Units (GLUs), which process the time-series data at each node to capture temporal evolution[27].

3.2 Local vs. Global Dependencies and Model Efficiency

Standard STGCN models often suffer from a strictly localized receptive field defined by the adjacency matrix. However, traffic patterns are deeply influenced by distant, non-adjacent zones. Advanced frameworks like LGSTGCN (Local-Global STGCN) and STGCN-MI integrate multi-head self-attention mechanisms alongside graph convolutions. The attention mechanism can dynamically learn correlation weights between all nodes in the network, allowing the model to capture critical global spatio-temporal dependencies that physical adjacency matrices ignore[24].

Furthermore, research into the computational efficiency of spatio-temporal models indicates that for strict short-term horizons (e.g., 10 minutes), a highly parameterized multi-block STGCN is often over-parameterized. A single-block STGCN (STGCN-1B) has been shown to match or exceed the accuracy of deeper models for short horizons while offering 60% higher throughput and reducing edge-device CPU latency by 38%[28]. This aligns perfectly with the

DSCI-41 KPI-6 requirement to maintain strict inference latency boundaries without degrading predictive quality[1].

3.3 Empirical Forecasting: Insights from the Zindi Challenge

The theoretical superiority of deep graph networks is frequently challenged when applied to noisy, real-world visual data. Insights from the Zindi Barbados Traffic Analysis Challenge provide profound operational context for the DSCI-41 O4 objective. The challenge tasked engineers with predicting congestion ratings (free flowing, light delay, moderate delay, heavy delay) at a roundabout 15 minutes into the future based on raw video streams from four cameras[29]. Competitors were restricted by a two-minute embargo to simulate real-world processing latency and explicitly barred from using future data during inference, effectively enforcing the DSCI-41 Negative Test NT-5 (prevention of information leakage across partitions)[1].

Competitors extracted highly diverse features from YOLO and ByteTrack pipelines, including flow rate, average vehicle counts, stopped vehicle detection, intensity motion, and deep learning scene embeddings via EfficientNet architectures[30]. A critical third-order insight from the challenge outcomes is the performance dichotomy between complex sequence models (LSTMs) and tree-based ensemble models (LightGBM, XGBoost, CatBoost). While LSTMs showed strong local validation performance when fed highly granular video-derived features (e.g., standard deviations of bounding box counts across frames), they failed completely to generalize to the hidden test set[31]. The noise introduced by upstream computer vision errors—such as missed detections due to occlusion or ID switches in tracking—compounded sequentially in the high-capacity deep learning models, leading to severe overfitting.

Conversely, gradient boosting models like LightGBM generalized exceptionally well. By aggregating the visual metrics into robust tabular temporal signals and applying strict 15-minute gap-free sliding windows, tree-based models ignored granular frame-by-frame visual noise and focused on macro-trends. The models optimized for the competition's specific metric (a weighted combination of 70% Macro-F1 and 30% Accuracy) demonstrated that in end-to-end visual forecasting pipelines, the robustness of the forecasting algorithm to upstream perceptual noise is far more critical than its theoretical capacity for complex sequence modeling[30].

Forecasting Methodology	Core Mechanism	Real-World Empirical Performance (Zindi Challenge)
LSTM / Sequence Models	Recurrent neural networks processing sequential feature vectors.	Prone to severe overfitting; highly sensitive to noise introduced by upstream visual detection errors.
STGCN / Graph Models	Graph convolutions paired	Excellent for city-scale

	with gated temporal units.	multi-node networks; mathematically sophisticated but overkill for single-intersection analysis.
Tree-Based Ensembles (LightGBM)	Gradient boosted decision trees utilizing aggregated sliding-window features.	Superior generalization on unseen test data; highly robust against noisy upstream computer vision signals.

4. Assurance, Validation, and Independent Acceptance

The final phase of the DSCI-41 capstone (O5) requires the validation of counting and forecasting against reference counts across varying conditions, culminating in an independent acceptance review[1]. The system must pass mandatory negative tests to ensure safe degradation, such as detecting camera calibration failures (NT-3) or average accuracy hiding critical misses on small targets (NT-4)[1].

To rigorously benchmark these capabilities, the literature relies on standardized datasets. The UA-DETRAC dataset, containing over 140,000 frames from fixed CCTV cameras, serves as the gold standard for evaluating multi-object tracking and vehicle counting under variable lighting[11]. For aerial and severe scale-variation testing, the VisDrone dataset provides exhaustive edge-case challenges[11].

Moreover, the integration of advanced trajectory similarity metrics—such as Cosine Similarity, Jensen-Shannon Divergence, and Euclidean Distance Similarity—enables the system to conduct comprehensive evaluations of vehicle movements, automatically flagging anomalous driving behaviors that deviate from normative traffic flow[16]. As the field progresses toward complex semantic understanding, datasets like NVIDIA's PhysicalAI Traffic Anomaly Reasoning (TAR) benchmark push beyond mere counting. Utilizing Vision-Language Models (VLMs) and multi-scale spatio-temporal event descriptions (MSTED), systems are now evaluated on their ability to perform chain-of-thought reasoning regarding driver decision-making, pedestrian intent, and causal analysis, measured through strict Mean Temporal IoU and Macro-F1 metrics[33].

5. Conclusion

The engineering mandate to transform raw roadside video into precise traffic volume estimates and short-term congestion forecasts requires a meticulously integrated, multi-disciplinary approach. Analysis of the current literature reveals that the evolution of single-stage detectors,

culminating in architectures like YOLOv11 and RT-DETR, provides the baseline visual intelligence required for mixed vehicle environments. However, sustaining accuracy under the degraded conditions mandated by the DSCI-41 specification requires abandoning computationally heavy image reconstruction in favor of illumination-aware network heads and advanced, low-confidence association trackers like ByteTrack.

Furthermore, translating pixel-based tracking into physical traffic flow requires overcoming the limitations of planar homography. The emerging integration of three-dimensional Look-Up Tables derived from external mapping systems represents a significant leap forward in geometric spatial awareness. Finally, the leap from historical volumes to future congestion states exposes a critical tension between theoretical graph modeling and empirical robustness. As demonstrated by real-world implementations, highly tuned tree-based models utilizing strict temporal sliding windows exhibit superior generalization by mathematically smoothing upstream perceptual noise. Ultimately, the successful deployment of this visual intelligence system relies on the seamless fusion of culturally adaptive perception, calibration-free spatial transformation, and noise-tolerant temporal forecasting.

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